FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014: 2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE(Main)-2015

ANSWERS, HINTS & SOLUTIONS

PART TEST – IV
(Main)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	A	B	B
2.	B	C	B
3.	C	C	B
4.	D	D	B
5.	C	C	D
6.	D	B	C
7.	C	A	B
8.	B	B	C
9.	A	A	C
10.	A	C	B
11.	D	B	D
12.	D	A	D
13.	A	B	B
14.	D	A	D
15.	A	D	B
16.	D	D	A
17.	B	C	A
18.	D	C	C
19.	B	A	C
20.	C	B	B
21.	A	B	D
22.	C	B	D
23.	C	A	C
24.	A	D	A
25.	D	B	A
26.	C	B	D
27.	A	C	D
28.	D	A	B
29.	D	B	A
30.	A	A	A

Physics

PART – I

1. $\rho = \dfrac{3\cdot14\times10^{-6}\times5}{1} = 15.7\mu$ ohm metre.

$\delta\rho = \rho\left[\dfrac{2\delta r}{r} + \dfrac{\delta L}{L} + \dfrac{\delta R}{R}\right]$ ohm metre.

$= 15\cdot7\mu\left[2\times\dfrac{\cdot01}{1} + \dfrac{\cdot01}{1} + \dfrac{\cdot01}{5}\right]$ ohm metre

$= 15.7\mu[\cdot02 + \cdot01 + \cdot002]$ ohm – metre

$= 15.7\times\cdot032\mu\,\Omega$ m

= 0.50 $\mu\Omega$m

2. $100\times\dfrac{\Delta P}{P} = 3\times0\cdot1 + 2\times0\cdot2 + 3\times0\cdot3 + 4\times0\cdot1$

$= \cdot3 + \cdot4 + \cdot9 + \cdot4 = 2\%$

3. $A = \lambda N_0 e^{-\lambda t_0}$

$3Ae = 3\lambda N_0 e^{-\lambda t}$

$\Rightarrow \dfrac{1}{e} = e^{\lambda(t-t_0)} = e^{-1}$

$t = t_0 - \dfrac{1}{\lambda}$.

4. $f = a^2\left(Z-b\right)^2$.

$\therefore$ (D)

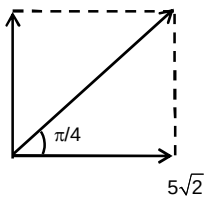
5. $\dfrac{I_{max}}{I_{min}} = \dfrac{\left(\sqrt{I_1}+\sqrt{I_2}\right)^2}{\left(\sqrt{I_1}-\sqrt{I_2}\right)^2} = \alpha$

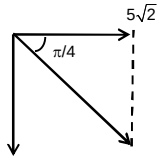
$\dfrac{I_1}{I_2} = \left(\dfrac{\sqrt{\alpha}+1}{\sqrt{\alpha}-1}\right)^2$ if $I_1 > I_2$

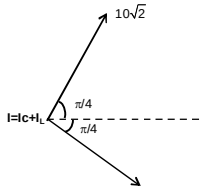
6. $I_c = \dfrac{100}{5\sqrt{2}}\sin\left(50t + \dfrac{\pi}{4}\right)$

$I_L = \dfrac{100}{5\sqrt{2}}\sin\left[50t - \dfrac{\pi}{4}\right]$

$I = 20\sin(50t)$







8. When u decreases v increases. For plane mirror u decreases.
$\therefore$ the image in plane mirror moves towards the left.

9. $n\lambda = 2\pi r_n$ and $r_n = n^2 r_1$

$\therefore \lambda : r_1 = 2\pi n : 1$

10. Apparent depth = $\frac{t_1}{\mu_1}+\frac{t_2}{\mu_2}$

Shift = $\left(t_1+t_2\right)-\left(\frac{t_1}{\mu_1}+\frac{t_2}{\mu_2}\right)$

$3.0=\left(t_1+10\right)-\left(\frac{t_1}{1.5}+\frac{10\times3}{4}\right)$

$3.0=\frac{t_1}{3}+2.5$

t_1 = 1.5 cm

11. In case of X-rays characteristic intensity of K_α, K_β, K_γ . . . . goes on decreasing while photon energy increases. This indicates that probabilities of P_1, P_2 and P_3 should be in increasing order.

12. Maximum energy of X – ray photon = 2000 eV
KE_{max} = 2000 – (100 + 5.8) = 1894.2 eV

13. $\frac{v_0}{v_i}=\frac{x^2}{y^2}$

x = 2 y

$\therefore \frac{1}{v}-\frac{1}{u}=\frac{1}{f}$

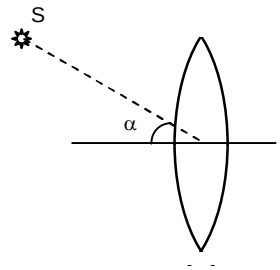
$\frac{1}{y}-\frac{1}{2y}=\frac{1}{f}$

$\frac{f}{y}=2$

15. $\Delta x=d\sin\alpha-d\frac{y}{D}=0$

$\Rightarrow$ y = D sin α

$=1000\times\frac{1}{200}=5$ mm. below.



16. m = 1 and $V_{xo} = V_{oy}$
$V_{xl} = m^2 V_{xo}$
$V_{yl} = -m V_{yo}$

$\therefore \theta=-\frac{\pi}{4}$

17. $\frac{\mu_2}{v'}-\frac{\mu_1}{4}=\frac{\mu_2-\mu_1}{R}$

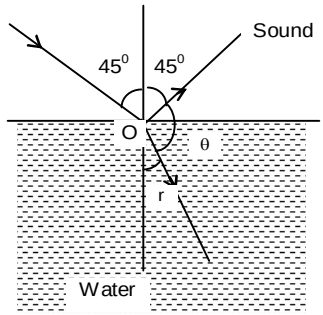
$\frac{\mu_3}{v}-\frac{\mu_2}{v'}=\frac{\mu_3-\mu_2}{-R}$ $\quad\Rightarrow\frac{\mu_3}{v}-\frac{\mu_1}{u}=\frac{\mu_3+\mu_1-2\mu_2}{R}$

$\frac{1.3}{x}-\frac{1}{-20}=\frac{\mu_3+\mu_1-2\mu_2}{R}$ $\quad\Rightarrow\frac{1}{y}-\frac{1.3}{-20}=\frac{\mu_3+\mu_1-2\mu_2}{R}$

Hence, $\frac{1.3}{x}+\frac{1}{20}=\frac{1}{y}+\frac{1.3}{20}$

$\therefore$ 1.3 y > x

18. The sound wave will be totally internally reflected on the surface.



For light wave

$1 \sin 45^0 = \frac{3\times10^8}{2.2\times10^8}\sin r$

$\Rightarrow r = \sin^{-1}\left(\frac{0.75}{\sqrt{2}}\right)\text{rad}.$

$\therefore \quad \theta = \pi - \frac{\pi}{4} - \sin^{-1}\left(\frac{0.75}{\sqrt{2}}\right) = \left[\frac{3\pi}{4} - \sin^{-1}\left(\frac{0.75}{\sqrt{2}}\right)\right]\text{rad}.$

19. $15^2 + 20^2 = 5^2 + X_C^2 \times 100 \Rightarrow X_C^2\, 100 = 600 \Rightarrow X_C = \sqrt{6}\ \Omega$

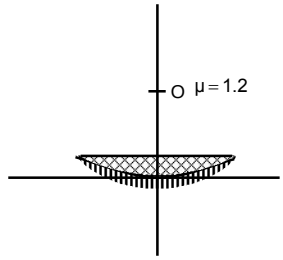
20. $F_T = F_m + mg = ILB - mg$

so, $I = \frac{E - B\ell V}{R} = \left(\frac{E - B\ell V}{R}\right)\ell B - mg = 0$

$E = \frac{mg}{R\ell B} + B\ell V$

21. Using symmetry, current will not flow in the resistor.

22. $\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{2}{f_2} = \frac{1}{15} + \frac{2\left({}^m\mu_\ell - 1\right)}{R_1} = \frac{1}{15} + \frac{{}^m\mu_\ell - 1}{15} = \frac{{}^m\mu_\ell}{15} \Rightarrow f_{eq} = \frac{15}{{}^m\mu_\ell}$



and $f_{eq} = \frac{(30 - 7.5)}{2} = \frac{22.5}{2}$

This $\frac{22.5}{2} = \frac{15}{{}^m\mu_1} \Rightarrow {}^m\mu_1 = \frac{4}{3}.$

Also ${}^m\mu_\ell = \frac{\mu_\ell}{\mu_m} \Rightarrow \frac{4}{3} = \frac{\mu}{1.2} \Rightarrow \mu = \frac{4\times1.2}{3} = 1.6$

23. For image to be coincident, either the rays should retrace or the image due to the lens should formed just at the pole of the mirror in thin case. The image formed due to lens is at 30 cm (2f) be from the lens. Thus either this image should be at centre of curvature of the convex mirror or at the pole of the mirror. Hence 6cm or 30 cm should be the separation between the lens and the mirror.

24. The charges on each capacitor will be equal to Q/2 (how?) when the current in the circuit is maximum.

29. Initially the current in the circuit is zero, and there is no potential drop across the internal resistance.

$L_1\frac{di_1}{dt} = L_2\frac{di_2}{dt} = \varepsilon, \text{ at } t = 0.$

At any time t,

$L_1\frac{di_1}{dt} = L_2\frac{di_2}{dt} = \varepsilon - ir$

30. $(\mu - 1)t = n\beta$

$\frac{(\mu_1 - 1)\times1.8\times10^{-5}}{(\mu_2 - 1)\times3.6\times10^{-5}} = \frac{18\beta}{9\beta}$

$(\mu_1 - 1) = 4(\mu_2 - 1)$

$4\mu_2 - \mu_1 = 3$

Chemistry

PART – II

SECTION – A

1. In fcc $a_2 = 2\sqrt{2}r$

$$\text{bcc } a_1 = \frac{4r}{\sqrt{3}}$$

$$\therefore \frac{a_1}{a_2} = \frac{4r}{\sqrt{3}} \times \frac{1}{2\sqrt{2}r} = \frac{\sqrt{2}}{\sqrt{3}}$$

2. In $K_3\left[Fe(C_2O_4)_3\right]$; O.N. of Fe = 3 + x + 3(–2) = 0

x = +3

$C_2O_4^{2-}$ is a bidentate ligand.

C.N of Fe = 3(2) = 6

3. $$4NaCl + K_2Cr_2O_7 + 3H_2SO_4 \longrightarrow 2CrO_2Cl_2 + 2Na_2SO_4 + K_2SO_4 + 3H_2O$$

$$Cr_2O_2Cl_2 + 4NaOH \longrightarrow Na_2CrO_4 + 2NaCl + 2H_2O$$

4. Cell reaction is $Zn + Cu^{2+} \longrightarrow Zn^{2+} + Cu$

$$E_{cell} = E^0_{cell} - \frac{RT}{nF} \ln \frac{\left[Zn^{2+}\right]}{\left[Cu^{2+}\right]}$$

Greater the factor $\left[Zn^{2+}\right]/\left[Cu^{2+}\right]$, less is the EMF.

6. He has the weakest vander waal's forces of attraction due to its smallest size and hence lowest boiling point. In other words, its liquefaction is difficult. As the size of noble gas increases from He to Xe, vander waal's forces increase and consequently boiling point increases and liquefaction becomes easy.

8. $$P^0_A = x_A P^0_A \qquad P^0_B = X_B P^0_B$$

$$y_A = \frac{P_A}{P_A + P_B} = \frac{x_A P^0_A}{x_A P^0_A + x_B P^0_B} = \frac{x_A P^0_A}{x_A P^0_A + (1 - x_A) P^0_B}$$

$$\frac{1}{y_A} = \frac{x_A\left(P^0_A - P^0_B\right) + P^0_B}{x_A P^0_A}$$

$$\frac{1}{x_A} = \frac{P^0_A}{P^0_b}\frac{1}{y_A} + \frac{P^0_B - P^0_A}{P^0_B}$$

$$\therefore \text{slope} = \frac{P^0_A}{P^0_B} \quad \text{Iintercept} = \frac{P^0_B - P^0_A}{P^0_B}$$

9. $$\left[Ag^+\right]_R = 8 \times 10^{-13} \quad \left[Ag^+\right]_L = 1.0M$$

$$E = 0 - 0.059 \log \frac{\left[Ag^+\right]_L}{\left[Ag^+\right]_R} = -0.71\,V$$

10. $\Delta G^0 = -nFE^0$
$\Delta H^0 - T\Delta S^0 = -nFE^0$
$E^0 = \frac{\Delta S^0}{nF}T - \frac{\Delta H^0}{nF}$
Solving for E^0 at 25°C and 20°C gives
$\Delta S^0 = -30.88\ JK^{-1}$

12. $Cl_2 + 2KOH \longrightarrow KCl + KClO + H_2O$ ...(i)
$3KClO \longrightarrow 2KCl + KClO_3$ ...(ii)
$4KClO_3 \longrightarrow 3KClO_4 + KCl$ ...(iii)
Multiply equation (i) by 12 and equation (ii) by 4 and add equation (i), (ii) and (iii) to obtain
$12Cl_2 + 24KOH \longrightarrow 21KCl + 3KClO_4 + 12H_2O$
852 415.5
1000 g $KClO_4$ will be produced from = $\frac{852 \times 1000}{415.5}$ g
= 2.05 kg

13. Isotonic solutions have same molar concentration.
6g urea/litre = 0.1 M
34.2 g sucrose/litre = 0.1 M

14. As k_1 is lowest, hence $A \xrightarrow{k_1} B$ is slowest.

16. $\left[Co(en)_2 Cl_2\right]Br$ shows ionization isomerism, cis and trans isomerism and optical isomerism.

17. Na_3VO_4 contains vanadium in +5 oxidation state which has vacant d – orbitals and is thus colourless.

18. $\Delta G° = -2FE° = -RT \ln K$
$K = 10^{10}$

19. $m = ZQ$; $m = \frac{108}{1 \times 96500} \times 965$ = **1.08**

20. For $Zn + 2Fe^{+2} \rightarrow Zn^{+2} + 2Fe$ $E^o_{cell} = -0.44 + 0.76 = +ve$ and hence, the reaction is spontaneous.

21. Lowering, $\Delta P \propto$ 'a' number of solute particles.
Hence, HA_1 is stronger acid than HA_2 (Given $\Delta P_1 > \Delta P_2$) and $pK_{a_1} < pK_{a_2}$.

23. $m = \frac{(639.7 - 631.9)}{639.7} \times \frac{1000}{78}$ = **0.156**.

24. The stabilization of the dispersed phase is due to adsorption of the ions.

25. Auto catalyst

30. In case of Ti, it is in +4 oxidation state and has no d-electrons in case of Cu d-orbitals are filled so in both cases d-d-transition will not take place and compound will be colourless.

Mathematics PART – III

1. Because $|\alpha\beta\gamma| = 15a\left(\left(a+\frac{31}{30}\right)^2+\frac{1259}{900}\right)=0$

Only if a = 0

2. $\frac{dy}{dx}+y^3(1+\cos x)-\frac{y}{x}=0$

$\frac{1}{y^3}\frac{dy}{dx}-\frac{1}{y^2x}=-(1+\cos x)$

Let $\frac{-1}{y^2}=u$

$\therefore \frac{1}{2}\frac{du}{dx}+\frac{u}{x}=-(1+\cos x)$

$IF = e^{\int\frac{2}{x}dx}=x^2$

$\therefore ux^2=-2\int x^2(1+\cos x)dx$

$\Rightarrow \frac{x^2}{2y^2}=\frac{x^3}{3}+x^2\sin x+2x\cos x-2\sin x+k$

3. $(\bar{p}\times\bar{q}+\bar{p})\times\bar{p}=\bar{q}$

$(\bar{p}\times\bar{q})\times\bar{p}=\bar{q}$

$\Rightarrow(\bar{p}.\bar{q})=0$

$\therefore [\bar{p}\ \ \bar{q}\ \ \bar{r}]=\bar{p}.(\bar{q}\times\bar{r})=\bar{p}.(\bar{q}\times(\bar{p}\times\bar{q}+\bar{p}))=(\bar{q}^2).(\bar{p}^2)=1$

4. $AA^T=I$

$\Rightarrow\begin{bmatrix}a&b&c\\c&a&b\\b&c&a\end{bmatrix}\begin{bmatrix}a&c&b\\b&a&c\\c&b&a\end{bmatrix}=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$

$\Rightarrow\begin{bmatrix}a^2+b^2+c^2&ac+ab+bc&ab+bc+ca\\ac+ab+bc&a^2+b^2+c^2&ab+bc+ca\\ab+bc+ac&bc+ac+ab&a^2+b^2+c^2\end{bmatrix}=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$

$a^2+b^2+c^2=1\Rightarrow ab+bc+ca=0$

$(a+b+c)^2=1+2(0)=1\Rightarrow a+b+c=\pm1$

6. $\begin{bmatrix}a_{11}&a_{12}\\a_{21}&a_{22}\end{bmatrix}\begin{bmatrix}1&1\\2&1\end{bmatrix}=\begin{bmatrix}1&1\\2&1\end{bmatrix}\begin{bmatrix}a_{11}&a_{12}\\a_{21}&a_{22}\end{bmatrix}$

$\Rightarrow a_{11}+2a_{12}=a_{11}+a_{21}$ ..... (i)

$a_{11}+a_{12}=a_{12}+a_{22}$ ..... (ii)

$a_{21}+2a_{22}=2a_{11}+a_{21}$

$a_{21}+a_{22}=2a_{12}+a_{22}$

$\therefore$ From (ii), $a_{11} = a_{22}$

$a_{21} = 2a_{12}$

$\because$ Given matrix is singular

$\Rightarrow a_{11}a_{22} = a_{21}a_{21}$

$\Rightarrow (a_{11})^2 = 2(a_{12})^2$

$\Rightarrow a_{11} = \sqrt{2}a_{12}$

7. AB = BA

8. $P = \frac{9}{25}$

9. Let $\bar{p} = p_1\hat{i} + p_2\hat{j} + p_3\hat{k}$

$\bar{p} \times \hat{i} = -p_2\hat{k} + p_3\hat{j}$

$\left|\bar{p} \times \hat{i}\right|^2 = p_2^2 + p_3^2$

$\left|\bar{p} \times \hat{j}\right|^2 = p_1^2 + p_3^2$

$\left|\bar{p} \times \hat{k}\right|^2 = p_1^2 + p_2^2$

Required solution = $2\left(p_1^2 + p_2^2 + p_3^2\right) = 2|\bar{p}|^2$

11. The lines should intersect or be parallel

Let $(2t+1, -t, 2t)$ be any point on the first line

Putting it in both the planes we get $\lambda = -5$

12. |adj adj adj A| = $|A|^{(n-1)^3} = 3^{16} \Rightarrow n = 3$

21. $P = \frac{2}{{}^8C_4} = \frac{2\times4\times3\times2\times1}{8\times7\times6\times5} = \frac{1}{35}$.

22. $\frac{-2xdx}{\sqrt{1-x^2}} == -2dy$

$\sqrt{1-x^2} = -(y+c)$

$x^2 + (y + c)^2 = 1$

$\therefore$ centre (0, – c).

23. $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} = a$

$\frac{x-4}{1} = \frac{y-6}{2} = \frac{z+k}{2} = b$

$x = 2a + 1, y = 3a + 2, z = 4a + 3$

$x = b + 4, y = 2b + 6, z = 2b – k$

$2a + 1 = b + 4$

$\therefore$ $2a – b = 3$ …(1)

$3a – 2b = 4$ …(2)

$a = 2, b = 1$

$4a + 3 = 2b - 12$

$\therefore k = 2b - 4a - 3 = 2 - 8 - 3 = -9.$

24. $\dfrac{\sin A}{a} = \dfrac{\sin B}{b} = \dfrac{\sin C}{c} = k$

$$\Delta = \begin{vmatrix} 1 & 4k & \cos A \\ 2a & 8ak & 1 \\ 3a & 12ak & \cos B \end{vmatrix} = 0.$$

25. a, b, c are in G.P. with common ratio r, such that $b = ar$, $c = ar^2$
Now system of equation have only zero (trivial) solution.

$$\text{If } \Delta = \begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix} \neq 0$$

$(a - b)(b - c)(c - a) \neq 0$
$(a - ar)(ar - ar^2)(ar^2 - a) \neq 0$
$a^3r(1 - r)^3(a + r) \neq 0$
$r \neq 0, \pm 1.$

26. $\lim\limits_{x\to 0} \tan\left(\dfrac{\pi}{4} + x\right)^{1/x} = e^2$

$\lim\limits_{x\to\infty} 1^x = 1$

$\lim\limits_{x\to 0} \dfrac{x^2 \sin\dfrac{1}{x}}{\sin x} = 0$

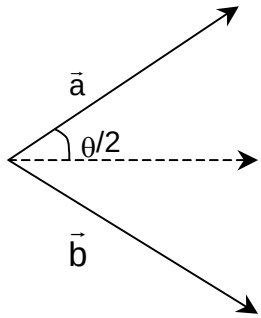
$\therefore \Delta = 0.$

27. $\left[\hat{a} \quad \vec{p} \quad \vec{q}\right]$ = projection of $\vec{p}\times\vec{q}$ in the direction of $\hat{a}$. Hence the given vector is $\vec{p}\times\vec{q}$

28. Vector in the direction of angular bisector of $\vec{a}$ and $\vec{b}$ $= \dfrac{\vec{a}+\vec{b}}{2}$

$\dfrac{\vec{a}+\vec{b}}{2}$ have magnitude $\cos(\theta/2)$

So, the unit vector in this direction will have magnitude $\dfrac{\vec{a}+\vec{b}}{2\cos\dfrac{\theta}{2}}$



29. If l, m, n be the d.c.'s of the line, them as it lies in both the given planes it is perpendicular to their normals i.e. $l - m + 2n = 0$ and $3l + m + n = 0$ or $\dfrac{l}{-3} = \dfrac{m}{5} = \dfrac{n}{4}$

$\therefore$ Equation of the line is $\dfrac{x-1}{-3} = \dfrac{y+2}{5} = \dfrac{z-3}{4}$

30. Required area is $\int\limits_{1/2}^{2} (2^x - \ln x)dx = \dfrac{3}{2} - \dfrac{5}{2}\ln 2 + \dfrac{4-\sqrt{2}}{\ln 2}$